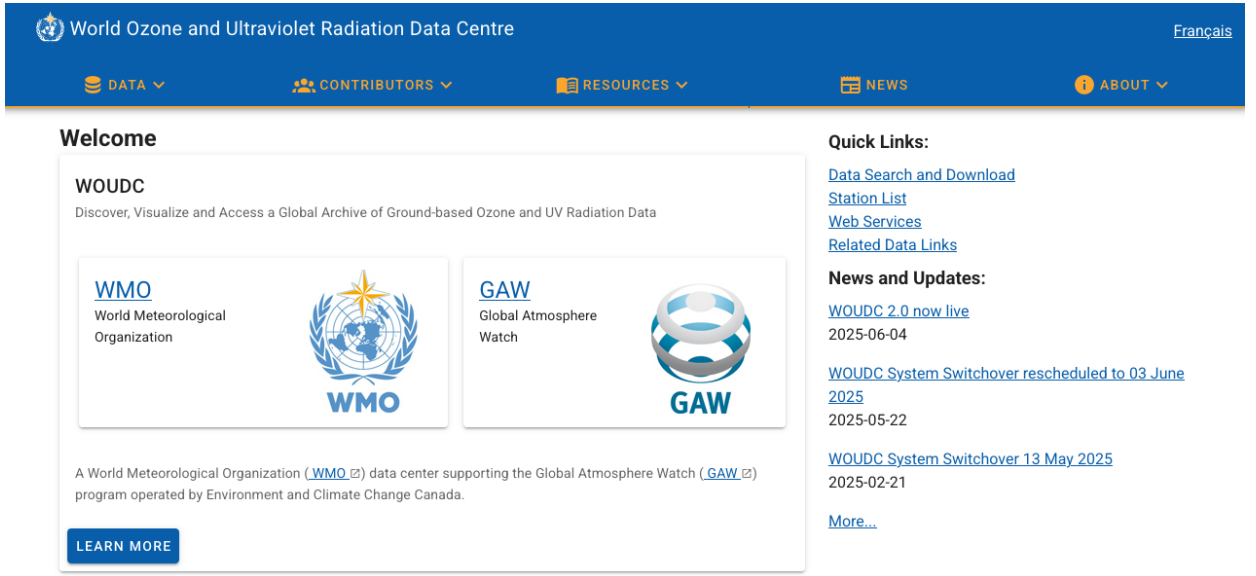
Environment and Climate Change Canada | Environnement et Changement climatique Canada

WMO WOUDC UPDATE

WMO SAG-O3-UV

2025-06-30



Tom Kralidis and Kevin Ngai
Geospatial and Open Data Systems
Meteorological Service of Canada
Environment and Climate Change Canada

Canada

WMO WOUDC UPDATE

- Data Centre status
- Standard operating procedures
- 2024 by the numbers
- WOUDC 2.0 updates

DATA CENTRE STATUS

- Meteorological Service of Canada
- Collaboration with ECCC Science and Technology Branch
- WMO reporting
  - SAG on Ozone/UV
  - WMO Expert Team on Atmospheric Composition Data Management
    - formerly ET on World Data Centres (ET-WDC)
  - Data Centre Interoperability (DCIO)

DATA CENTRE STATUS

- 2015: renewal (WOUDC 1.0)
- 2025: platform upgrade (WOUDC 2.0)
- Current operations
  - Contributor support
  - User support
  - Continuous data improvement
    - Data ingest
    - Metadata quality assessment / correction
  - Enhancements

DATA CENTRE STATUS

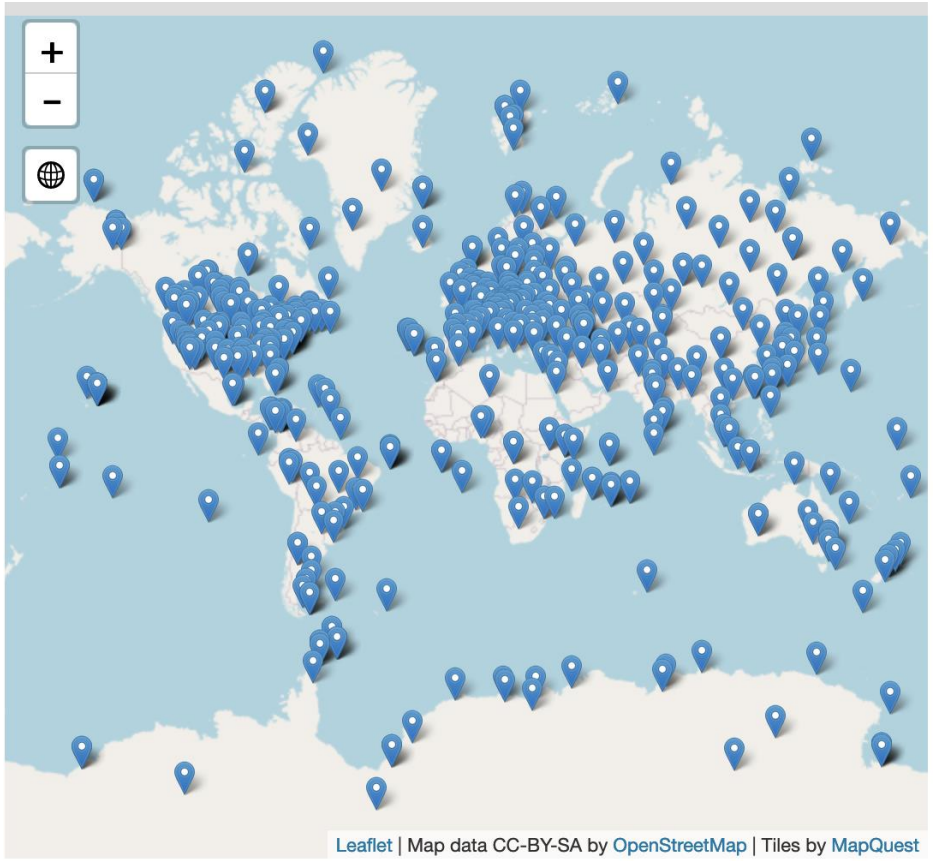
- 2024-02-01 to 2024-03-14: Metrics server issue resulted in missing logs
- WOUDC 2.0 implementation
- Complete data re-ingest, numerous corrections:
  - Metadata QC
  - Data files updates (moving to various folders)
  - Discarded empty data files with contributor's permission
    - Remaining empty files on the WAF are not ingested in WOUDC 2.0

STANDARD OPERATING PROCEDURES

- Submissions, contact validation, metadata QC ongoing, unchanged
  - See Annex for details
- WOUDC Guidebook: guide.woudc.org

NUMBER OF FILES BY DATASET (2025-06-27)

471 stations (duplicates removed)



Broad-band 1.0	172588
Broad-band 2.0	4084
OzoneSonde 1.0 (form 1)	100807
OzoneSonde 1.0 (form 2)	1638
Lidar	675
Multi-band	93778
TotalOzoneObs 1.0	167255
TotalOzone 1.0	91874
TotalOzone 2.0	494
Spectral 1.0	325286
Spectral 2.0	4050
UmkehrN14 1.0	10886
UmkehrN14 2.0	10240
RocketSonde	176
TOTAL	**983821**

NEW CONTRIBUTORS

- National Agency of Civil Aviation and Meteorology (Comoros)
- Hokkaido University (Japan)
- Center of Atmospheric Watch and service, China Meteorological Administration (China)
- Yunnan University (China)
- Center of Excellence Telesensing of Environment and Model Prediction of Severe Events - University of L'Aquila (Italy)
- Uruguayan Institute of Meteorology (Uruguay)

NEW STATIONS

- Moroni
- Akedala
- Shangdianzi
- Kunming
- Casale Calore
- Bharati
- Hokkaido Sapporo University
- Salto Grande [returned]

PROCESSING RUNS 2024

Month	Total Files	Passed	Fixed	Failed (extCSV)	Failed (non-standard)	% Passed	% Passed With Fixes	% Failed
January	1567	1060	351	499	338	67.65	22.40	53.41
February	1189	768	296	421	57	64.59	24.89	40.20
March	1859	1538	293	326	125	82.73	15.76	24.26
April	3203	2517	21	573	14	78.58	0.66	18.33
May	5778	5392	215	386	32	93.32	3.72	7.23
June	3146	2825	819	284	1	89.80	26.03	9.06
July	4056	2159	665	1885	1859	53.23	16.40	92.31
August	1956	735	53	1221	4	37.58	2.71	62.63
September	1328	1312	38	16	0	98.80	2.86	1.20
October	3433	964	365	2468	351	28.08	10.63	82.11
November	2836	2685	134	0	0	94.68	4.72	0.00
December	5027	4328	80	697	0	86.10	1.59	13.87
Total	**35378**	**26283**	**3330**	**8776**	**2781**	**74.29**	**9.41**	**32.67**

PROCESSING RUNS 2024

- Issues noted
  - Hokkaido University: new contributor, duplicate files
  - MeteoSwiss: publishing to 2 agencies
- Removing these issues and duplicate file publishing results in 14.87 % Failed (last column)

PROCESSING RUNS BY YEAR

Year	Total Files	Passed	Fixed	Failed (extCSV)	Failed (non-standard)	% Passed	% Passed With Fixes	% Failed
2016	34451	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2017	44022	28049	6863	2367	6743	63.72	15.59	20.69
2018	31224	19364	3496	1704	6560	62.02	11.20	26.47
2019	112249	98871	12121	927	330	88.08	10.80	1.12
2020	100662	73046	24439	2071	1056	72.57	24.28	3.11
2021	32529	21086	10199	897	615	64.82	31.35	4.65
2022	22192	22127	2317	1626	626	99.71	10.44	10.15
2023	50953	32813	2015	3399	491	64.40	3.95	7.63
2024	35378	26283	3330	8776	2781	74.29	9.41	32.67

DOWNLOADS 2024

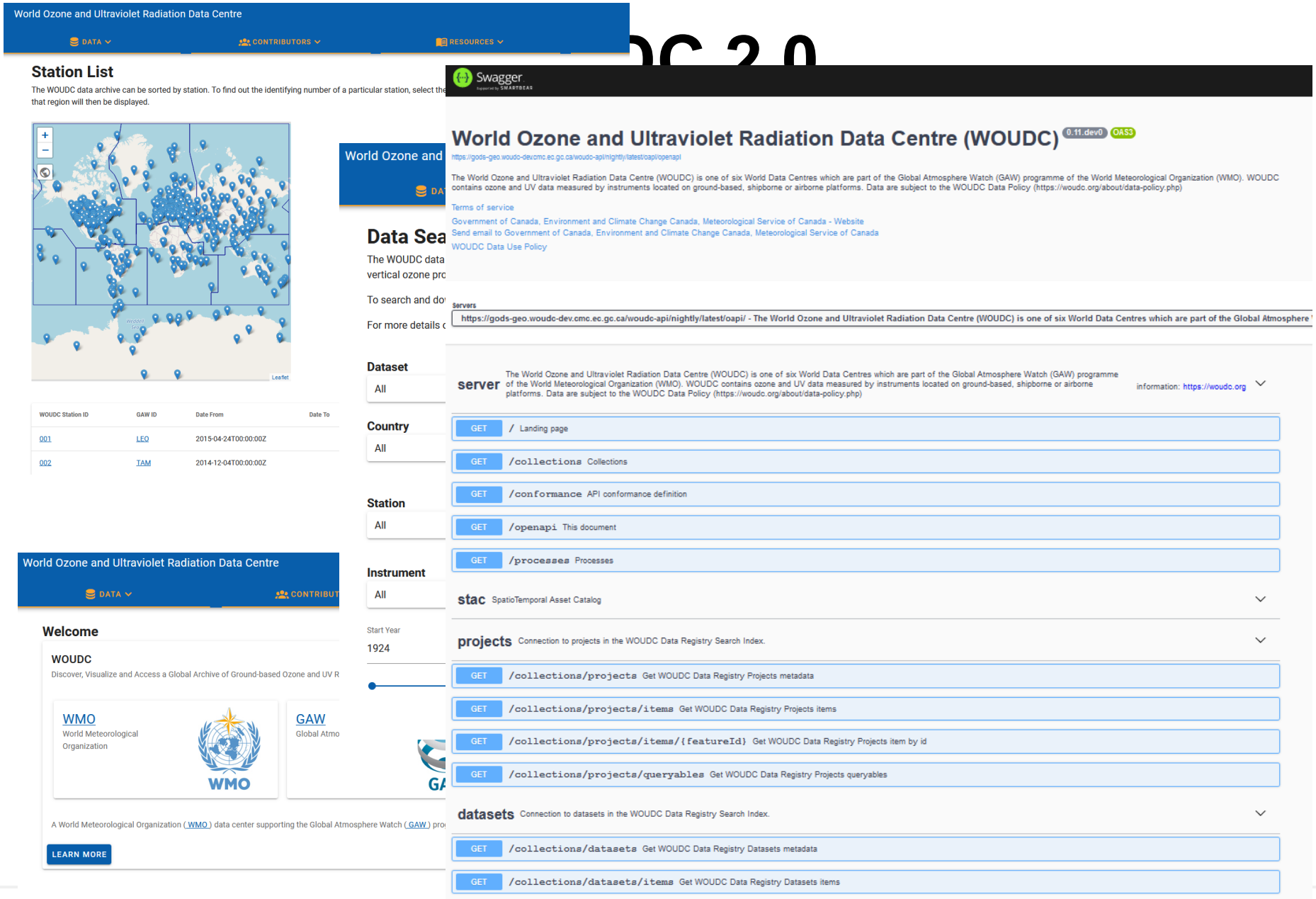
(excludes 2024-02-01 to 2024-03-14)

Dataset	WAF (number of downloads)
OzoneSonde_1.0_1	25606035
TotalOzone_1.0_1	18754888
TotalOzoneObs_1.0_1	8685276
Spectral_1.0_1	3760790
Broad-band_1.0_1	1817432
Multi-band_1.0_1	1019017
UmkehrN14_2.0_1	305962
UmkehrN14_1.0_1	304584
Broad-band_2.0_1	47489
Spectral_2.0_1	43168
RocketSonde	12148
TotalOzone_2.0_1	33393
Lidar_1.0_1	16394
SurfaceOzone_1.0_1	44759
Total	**60451335**

Dataset	Geospatial Web Services (number of requests)
OzoneSonde	16239
TotalOzone	10240
uv_index_hourly (data product)	7415
TotalOzoneObs	4797
RocketSonde	2926
Broadband	2856
Spectral	2638
Lidar	2406
Multiband	2189
UmkehrN14 1.0	810
UmkehrN14 2.0	610
Total	**53126**

WOUDC 2.0 RELEASE

- Simplified data management
- New UI: woudc.org
- New API: api.woudc.org
- Notification services
- All software is free and open source on GitHub
- WIS2 implementation
- Dobson cross-section correction in progress (Xiaoyi)


World Ozone and Ultraviolet Radiation Data Centre
DATA
CONTRIBUTORS
RESOURCES
Station List
The WOUDC data archive can be sorted by station. To find out the identifying number of a particular station, select the
that region will then be displayed.
Leaflet
WOUDC Station ID
GAW ID
Date From
Date To
001
LEQ
2015-04-24T00:00:00Z
002
TAM
2014-12-04T00:00:00Z
World Ozone and
Data Sea
The WOUDC data
vertical ozone pro
To search and do
For more details
Dataset
All
Country
All
Station
All
Instrument
All
Start Year
1924
World Ozone and Ultraviolet Radiation Data Centre
DATA
CONTRIBUT
Welcome
WOUDC
Discover, Visualize and Access a Global Archive of Ground-based Ozone and UV R
WMO
World Meteorological
Organization
WMO
GAW
Global Atmo
A World Meteorological Organization (WMO) data center supporting the Global Atmosphere Watch (GAW) pro
LEARN MORE
Swagger
World Ozone and Ultraviolet Radiation Data Centre (WOUDC)
0.11.dev0
OAS3
https://gods-geo.woudc-dev.cmc.ec.gc.ca/woudc-api/nightly/latest/oapi/openapi
The World Ozone and Ultraviolet Radiation Data Centre (WOUDC) is one of six World Data Centres which are part of the Global Atmosphere Watch (GAW) programme of the World Meteorological Organization (WMO). WOUDC contains ozone and UV data measured by instruments located on ground-based, shipborne or airborne platforms. Data are subject to the WOUDC Data Policy (https://woudc.org/about/data-policy.php)
Terms of service
Government of Canada, Environment and Climate Change Canada, Meteorological Service of Canada - Website
Send email to Government of Canada, Environment and Climate Change Canada, Meteorological Service of Canada
WOUDC Data Use Policy
Servers
https://gods-geo.woudc-dev.cmc.ec.gc.ca/woudc-api/nightly/latest/oapi/ - The World Ozone and Ultraviolet Radiation Data Centre (WOUDC) is one of six World Data Centres which are part of the Global Atmosphere
server
The World Ozone and Ultraviolet Radiation Data Centre (WOUDC) is one of six World Data Centres which are part of the Global Atmosphere Watch (GAW) programme of the World Meteorological Organization (WMO). WOUDC contains ozone and UV data measured by instruments located on ground-based, shipborne or airborne platforms. Data are subject to the WOUDC Data Policy (https://woudc.org/about/data-policy.php)
information: https://woudc.org
GET / Landing page
GET /collections Collections
GET /conformance API conformance definition
GET /openapi This document
GET /processes Processes
stac SpatioTemporal Asset Catalog
projects Connection to projects in the WOUDC Data Registry Search Index.
GET /collections/projects Get WOUDC Data Registry Projects metadata
GET /collections/projects/items Get WOUDC Data Registry Projects items
GET /collections/projects/items/{featureId} Get WOUDC Data Registry Projects item by id
GET /collections/projects/queryables Get WOUDC Data Registry Projects queryables
datasets Connection to datasets in the WOUDC Data Registry Search Index.
GET /collections/datasets Get WOUDC Data Registry Datasets metadata
GET /collections/datasets/items Get WOUDC Data Registry Datasets items

Evolution of WMO data exchange

1963 World Weather Watch

1970s Global Telecommunication System (GTS)

2007 WMO Information System (WIS)

2019 WMO Reform (Earth System Approach)

2021 WMO Unified Data Policy (Core, Recommended)

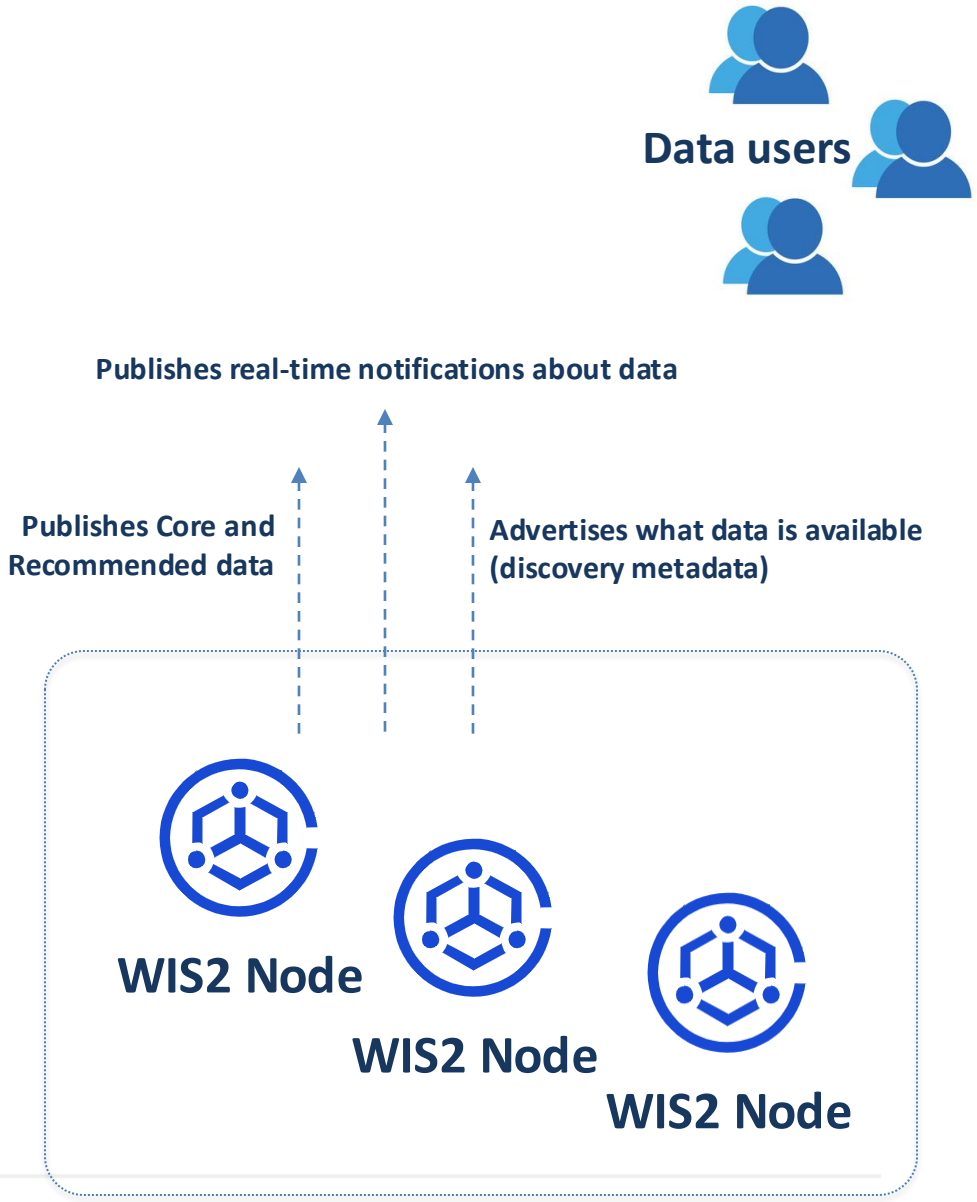
WIS 2.0

… collaborative system of systems using Web-architecture and open standards to provide simple, timely and seamless sharing of trusted data and information …

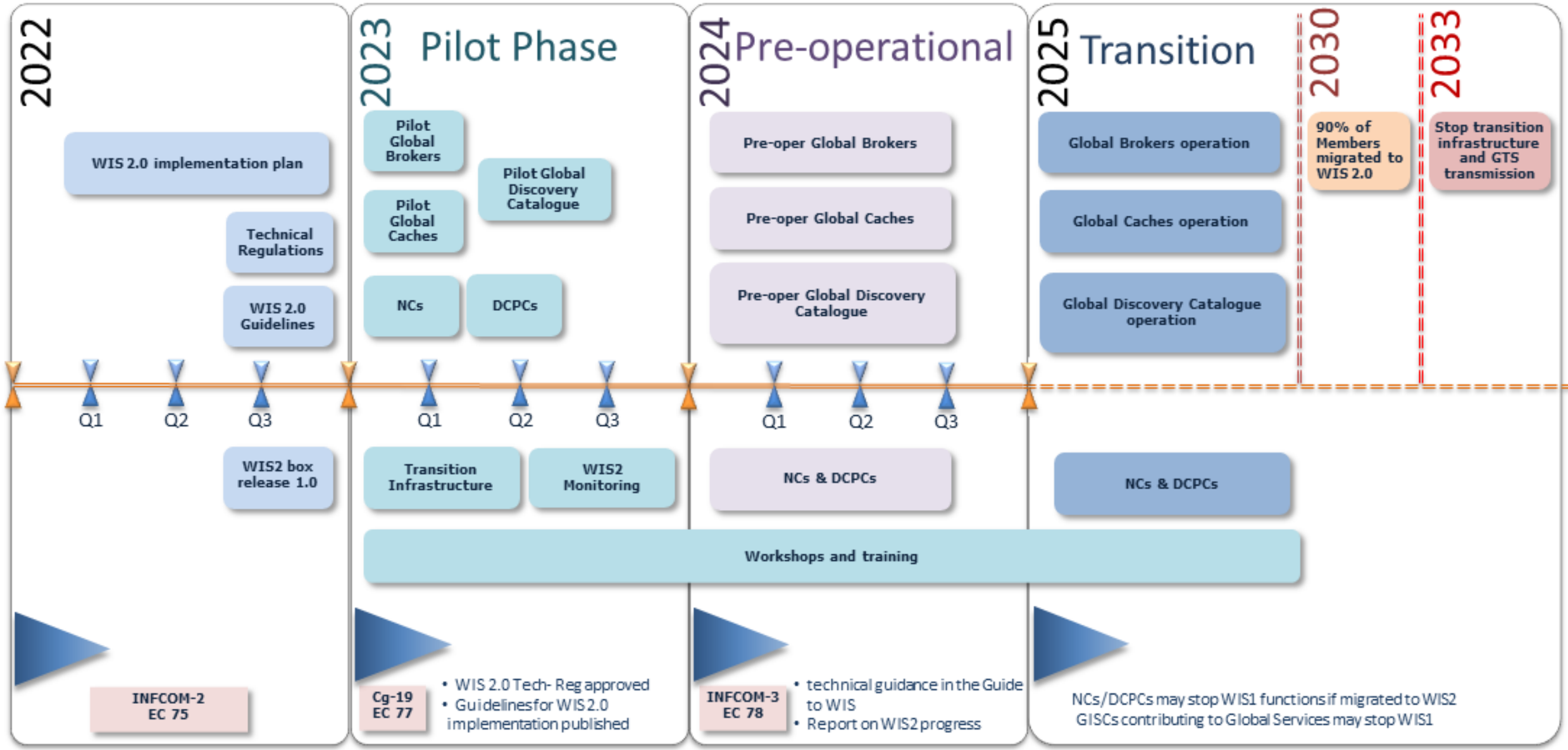
- Open Standards (OGC, W3C, IETF, …)
- Free and Open Source tooling
- Data sharing through Web and real-time notifications with publication/subscription (Pub/Sub) protocols
- Cloud ready (turn-key solutions)
- Web APIs (Application Programming Interface)

WIS2 Components: WIS2 Nodes

- Each WMO Member shall implement at least one WIS2 Node to share data in WIS2
- A WIS2 Node replaces the GTS Message Switching System
- Data and metadata are shared using a WIS2 Node
- A WIS2 Node shares data via an HTTPS service and sends notifications to MQTT subscribers

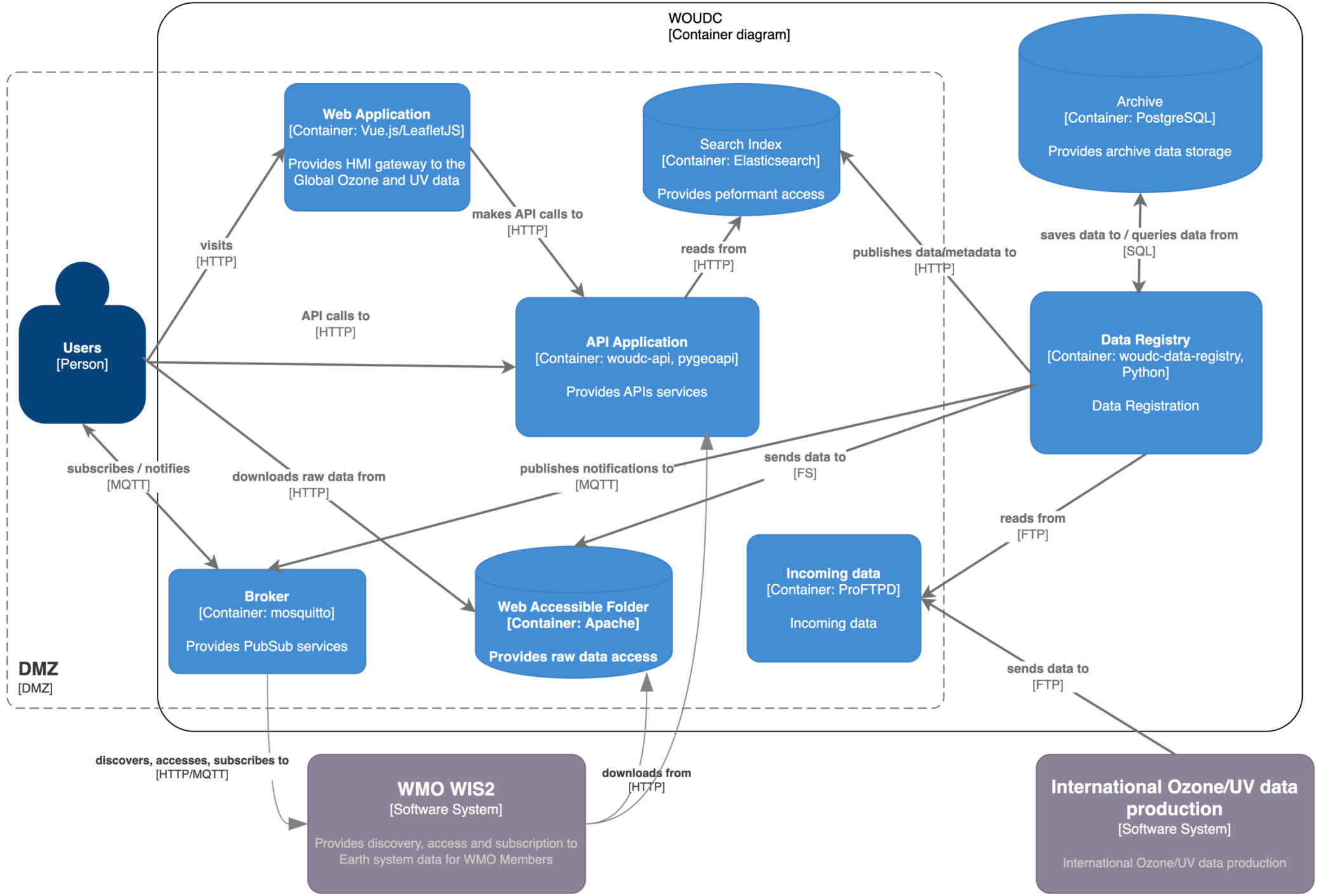


WIS2 Transition



WIS 2.0 IMPLEMENTATION

- WOUDC datasets will be discoverable on the WIS2 Catalogue (search)
- WOUDC data will be published and made available to WIS2 in real-time as part of the data processing/publication process
- Topic definitions for atmospheric composition data publication (ET-ACDM)
  - github.com/wmo-im/wis2-topic-hierarchy/issues/233


WOUDC
[Container diagram]
Web Application
[Container: Vue.js/LeafletJS]
Provides HMI gateway to the Global Ozone and UV data
Search Index
[Container: Elasticsearch]
Provides peformant access
Archive
[Container: PostgreSQL]
Provides archive data storage
visits
[HTTP]
makes API calls to
[HTTP]
reads from
[HTTP]
publishes data/metadata to
[HTTP]
saves data to / queries data from
[SQL]
Users
[Person]
API calls to
[HTTP]
API Application
[Container: woudc-api, pygeoapi]
Provides APIs services
Data Registry
[Container: woudc-data-registry, Python]
Data Registration
subscribes / notifies
[MQTT]
downloads raw data from
[HTTP]
publishes notifications to
[MQTT]
sends data to
[FS]
reads from
[FTP]
Broker
[Container: mosquitto]
Provides PubSub services
Web Accessible Folder
[Container: Apache]
Provides raw data access
Incoming data
[Container: ProFTPD]
Incoming data
DMZ
[DMZ]
sends data to
[FTP]
discovers, accesses, subscribes to
[HTTP/MQTT]
WMO WIS2
[Software System]
Provides discovery, access and subscription to Earth system data for WMO Members
downloads from
[HTTP]
International Ozone/UV data production
[Software System]
International Ozone/UV data production

WMO WOUDC UPDATE

Thank you!

- Website: woudc.org
- Guidebook: guide.woudc.org
- GitHub/Code: github.com/woudc
- Wiki: github.com/woudc/woudc/wiki

EXTERNAL DATA CENTRE LINKAGES

- European UV database: available via woudc.org/en/resources/links
- EUBREWNET
  - File indexing/discovery
- NDACC
  - File indexing/discovery
- SHADOZ
  - Full data conversion/ingest
    - Action for follow-up

STANDARD OPERATING PROCEDURES

CONTRIBUTOR CONTACT VALIDATION

- Annual notice sent to all contributors to validate their contact information

CONTRIBUTOR DATA SUBMISSION REMINDERS

- [1] Have contributed data within last 10 years
  - AND
- [2] No data received within the last 6 months
- Reminder notification sent to all contributors who meet both [1] and [2]
- The process runs monthly
- Contributor will be notified until they do **not** meet [2]
- 10 years denotes active contributors and stations
- Process can be configured as required; multiple messages, intervals between messages, sessions, cc list, bcc list, etc.
- Record of notifications archived

DATA SUBMISSION NOTIFICATIONS

- Implemented December 2017
- Contributors notified via email
  - Data Submission Confirmation (nightly)
    - List of submitted files to ftp.woudc.org contributor account
  - Data Processing report (weekly)
    - Total files received
    - Number of passed files
    - Number of manually repaired files
    - Number of failed files
    - Summary of Manual fixes

METADATA QUALITY CORRECTION

- Metadata qa/qc
- Manual fixing of metadata for data submissions (in parallel to contributor feedback)
- Bulk corrections (ongoing) in consultation with various stakeholders